1D Spin Chain `QuDPy` Simulations

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Abstract

Towards a PRL draft, Parallel `QuDPy` simulations of allowed orbital tranitions and their coupling to dark transitions.

GRAND QUESTION

Can the coherence and dephasing between optically bright and nominally dark states serve as a direct probe of many-body quantum correlations in a nonclassically ordered spin system?

HIGH-LEVEL SCIENTIFIC QUESTIONS

The experimental and modeling program is guided by a small set of high-impact, falsifiable questions that target the microscopic origin of the observed spectral correlations:

- **Are bright–dark cross peaks a signature of coherent eigenstate mixing?**

  Do cross peaks persist in the absence of dephasing and population transfer,

  $$\Gamma_\phi = 0, \qquad k_{B\to D} = 0, \tag{1}$$

  and vanish when the mixing term is removed,

  $$V_{BD} \to 0\,? \tag{2}$$

  This directly tests whether the nonlinear response accesses nominally dark sectors of the Hilbert space through coherent hybridization.

- **What microscopic quantity does the cross-peak amplitude measure?**

  Is the cross peak governed by

  $$S_{\text{cross}}(T, B) \propto |V_{BD}(T, B)|^2, \tag{3}$$

  and therefore a proxy for spin–orbital correlations, or does it instead track trivial spectral parameters such as energy splittings or oscillator strength?

- **Does magnetic field tune coherent mixing or only shift energies?**

  Does

$$\frac{\partial S_{\mathrm{cross}}}{\partial B} \neq 0 \tag{4}$$

  reflect a modification of the mixing matrix element $V_{BD}(T, B)$, or merely a field-induced shift of the diagonal resonances? This distinguishes a genuine many-body response from a single-particle effect.

- **Can nonlinear spectroscopy distinguish thermal from nonthermal states?**

  Following impulsive excitation, does the spectral response correspond to a quasithermal redistribution,

$$\rho \sim e^{-H/k_B(T+\Delta T)}, \tag{5}$$

  or to a nonthermal state in which the Hamiltonian itself is modified? This question is central to correlated systems, where interaction quenches can produce metastable nonthermal states rather than simple heating.

- **Do cross peaks encode short-range spin correlations?**

  Is it possible to establish a mapping

$$S_{\mathrm{cross}}(T, B) \longleftrightarrow C_1(T, B),\ \delta(T, B), \tag{6}$$

  where C_1 is the nearest-neighbor spin correlation and δ the dimerization order parameter? If so, the nonlinear optical response becomes a direct probe of nonclassical spin correlations in the absence of long-range order.

- **What is the role of detection in accessing dark states?**

  Are the observed correlations intrinsic to the system Hamiltonian, or are they selected by the detection observable,

$$\hat{O}_{\mathrm{det}}\ ? \tag{7}$$

  This addresses whether nonlinear spectroscopy measures universal properties or detection-dependent projections of many-body dynamics.

- **Are the observed correlations robust to experimental and numerical artifacts?**

  Do the cross peaks persist under variation of pulse bandwidth, temporal overlap, and numerical truncation? This establishes that the effect is not generated by finite-pulse or simulation artifacts but reflects intrinsic system dynamics.

MODEL AND QUDPY SIMULATION STRATEGY

Hamiltonian

We model a quasi-1D $S = 1/2$ Cu^{2+} spin chain with a spin–Peierls instability, coupled to a local orbital excitation that provides the optical handle. The total Hamiltonian is

$$H = H_{\text{spin}} + H_{\text{orb}} + H_{\text{spin-orb}}. \tag{8}$$

The spin sector is

$$H_{\text{spin}} = \sum_i J_i(T,B)\,\mathbf{S}_i \cdot \mathbf{S}_{i+1} + J_2 \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+2} + g\mu_B B \sum_i S_i^z, \tag{9}$$

with bond-alternating exchange

$$J_i(T,B) = J\left[1 + (-1)^i \delta(T,B)\right]. \tag{10}$$

The spin–Peierls order parameter is modeled as

$$\delta(T,B) = \delta_0 \left[1 - \frac{T}{T_{\text{SP}}(B)}\right]^\beta, \qquad T < T_{\text{SP}}(B), \tag{11}$$

and $\delta = 0$ otherwise.

The orbital (optical) sector is truncated to a bright–dark manifold:

$$H_{\text{orb}} = E_B |B\rangle\langle B| + E_D |D\rangle\langle D|. \tag{12}$$

Spin–orbital coupling induces bright–dark mixing:

$$H_{\text{spin-orb}} = V_{BD}(T,B)\left(|B\rangle\langle D| + |D\rangle\langle B|\right). \tag{13}$$

We parameterize the coupling as

$$V_{BD}(T,B) = V_0 + \lambda_\delta\,\delta(T,B) + \lambda_C\,C_1(T,B), \tag{14}$$

where

$$C_1(T,B) = \langle \mathbf{S}_i \cdot \mathbf{S}_{i+1} \rangle_{T,B}. \tag{15}$$

Light–Matter Interaction

The dipole operator is

$$\hat{\mu} = \mu_B |g\rangle\langle B| + \mu_D |g\rangle\langle D| + \text{h.c.} \tag{16}$$

with $\mu_D \ll \mu_B$ (nominally dark state).

Equilibrium State

The thermal density matrix is

$$\rho_{\text{eq}}(T, B) = \frac{e^{-H/k_B T}}{Z}. \tag{17}$$

Linear and Nonlinear Response

The linear absorption spectrum is

$$S^{(1)}(\omega; T, B) = \text{Im} \int_0^\infty dt\, e^{i\omega t} \text{Tr}\left[\hat{\mu}(t)\hat{\mu}(0)\rho_{\text{eq}}(T, B)\right]. \tag{18}$$

The third-order 2D coherent spectrum is

$$S^{(3)}(\omega_3, t_2, \omega_1; T, B), \tag{19}$$

computed from the standard Liouville pathway expansion.

Cross-Peak Observable

We define the bright–dark cross-peak amplitude as

$$S_{\text{cross}}(T, B) \equiv S^{(3)}(\omega_D, t_2, \omega_B; T, B). \tag{20}$$

In the coherent-mixing limit ($\Gamma_\phi = 0$, no population transfer),

$$S_{\text{cross}}(T, B) \propto |\mu_B|^2 \, |V_{BD}(T, B)|^2. \tag{21}$$

QuDPy Implementation Strategy

We implement the model in QuDPy using a reduced Hilbert space:

- Finite spin chain ($N = 6$–10) with exact diagonalization.
- Orbital subspace $\{|g\rangle, |B\rangle, |D\rangle\}$.
- Tensor-product construction of spin and orbital sectors.

The simulation protocol consists of controlled parameter sweeps:

(i) Coherent limit

$$\Gamma_\phi = 0, \qquad k_{B\to D} = 0. \tag{22}$$

(ii) Mixing control

$$V_{BD} \to 0. \tag{23}$$

(iii) Dark-state oscillator strength

$$\mu_D \to 0. \tag{24}$$

(iv) Temperature and field sweep

$$(T, B): \quad T \in [0, 2T_{\mathrm{SP}}], \quad B \in [0, B_{\max}]. \tag{25}$$

Key Diagnostic

The central test is: $\mathrm{S_{cross}} \neq 0 \quad \text{for} \quad \Gamma_\phi = 0,\ k_{B\to D} = 0,$
$S_{\mathrm{cross}} \to 0 \quad \text{as} \quad V_{BD} \to 0.$

This establishes that the cross peak arises from coherent bright–dark mixing rather than dephasing or population transfer.

Magnetic-Field Dependence

The field dependence of the cross peak follows

$$S_{\mathrm{cross}}(T, B) \propto |V_0 + \lambda_\delta \delta(T, B) + \lambda_C C_1(T, B)|^2. \tag{26}$$

Thus,

$$\frac{\partial S_{\mathrm{cross}}}{\partial B} \tag{27}$$

provides a direct spectroscopic probe of field-induced changes in short-range spin correlations and dimerization.

Minimal Simulation Strategy and Controls

At the current stage, the objective is not maximal microscopic completeness but the identification of the minimal physical ingredients required to reproduce the observed cross peaks. We therefore begin with a reduced bright–dark model:

$$H_{\mathrm{BD}} = (\ E\ )_B\, V_{BD}(T,B) V_{BD}(T,B) E_D. \tag{28}$$

The temperature- and field-dependent mixing is parameterized phenomenologically as

$$V_{BD}(T,B) = V_0 + \lambda_B B + \lambda_T f(T), \tag{29}$$

where $f(T)$ is a smooth function capturing thermal evolution (to be refined later in terms of $\delta(T,B)$ or $C_1(T,B)$).

Controlled simulation protocol. The QuDPy simulations are performed under a sequence of controlled limits designed to isolate the physical origin of the cross peaks:

(i) Coherent limit

$$\Gamma_\phi = 0, \qquad k_{B\to D} = 0, \tag{30}$$

which removes all bath-induced dephasing and population transfer.

(ii) Mixing suppression

$$V_{BD} \to 0, \tag{31}$$

which eliminates coherent bright–dark hybridization.

(iii) Dark-state oscillator strength suppression

$$\mu_D \to 0, \tag{32}$$

ensuring that the dark state is optically inactive at the one-photon level.

(iv) Inclusion of dephasing

$$\Gamma_\phi \neq 0, \tag{33}$$

introduced only after the coherent-limit behavior is established.

Diagnostic criterion. The central test of the model is:

$\mathrm{S}_{\text{cross}} \neq 0 \quad \text{for} \quad \Gamma_\phi = 0,\ k_{B\to D} = 0,$

$S_{\text{cross}} \to 0 \quad \text{as} \quad V_{BD} \to 0.$

This establishes that the cross peak originates from coherent bright–dark mixing, rather than dephasing or population transfer processes.

Magnetic-field discrimination. Magnetic field is treated as a diagnostic control parameter. The relevant observable is

$$\frac{\partial S_{\text{cross}}}{\partial B}. \tag{34}$$

A trivial spectral shift produces no change in cross-peak amplitude, whereas

$$S_{\text{cross}}(T,B) \propto |V_{BD}(T,B)|^2 \tag{35}$$

indicates that the field modulates coherent mixing.

Numerical observables. For each (T,B) point, we extract:

- Cross-peak amplitude: $S_{\text{cross}}(T,B)$,
- Antidiagonal linewidth: $\Gamma_{\text{cross}}(T,B)$,
- Peak positions: (ω_B, ω_D).

Numerical consistency checks. To exclude artifacts, we verify invariance of the cross peaks with respect to pulse bandwidth and temporal overlap, ensuring that the signal is not generated by finite pulse effects.

Model hierarchy. Only after establishing the minimal coherent-mixing mechanism do we upgrade the model to

$$V_{BD}(T,B) \longrightarrow V_0 + \lambda_\delta \delta(T,B) + \lambda_C C_1(T,B), \tag{36}$$

thereby connecting the nonlinear optical response to microscopic spin correlations.

MODEL HIERARCHY

The first modelling layer follows the simplified spin–orbital Hamiltonian of Wohlfeld *et al.*, in which the orbital excitation propagates in an antiferromagnetic spin background without explicitly including the spin–Peierls transition. This model is valuable because it isolates the intrinsic coupling between orbital excitations and spin fluctuations.

$$H = 2J \sum_{\langle i,j \rangle} \left(\mathbf{S}_i \cdot \mathbf{S}_j + \frac{1}{4} \right) \left(\mathbf{T}_i \cdot \mathbf{T}_j + \frac{1}{4} \right) + E_z \sum_i T_i^z. \tag{37}$$

Here $\mathbf{S}_i$ describes the Cu^{2+} spin degree of freedom and $\mathbf{T}_i$ describes the local orbital pseudospin. The term E_z represents the crystal-field splitting between orbital states.

This simplified Hamiltonian should be used first because it tests whether the observed bright–dark cross peaks can arise from intrinsic spin–orbital dynamics without invoking spin–Peierls dimerization, thermal disorder, or phenomenological dephasing.

The key diagnostic is whether the field-emission cross peak can be reproduced as a coherence between an optically bright orbital excitation and a nominally dark spin–orbital excitation:

$$S_{\text{cross}} \neq 0 \tag{38}$$

in the absence of phenomenological dephasing or population transfer.

Only after this baseline mechanism is established should the model be extended to include spin–Peierls physics through bond alternation,

$$J_i = J \left[1 + (-1)^i \delta(T, B) \right], \tag{39}$$

and magnetic-field tuning,

$$H_B = g\mu_B B \sum_i S_i^z. \tag{40}$$